

Mobile Tool Assistant

User Manual

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Introduction

Thank you for using the **Mobile Tool Assistant**. This robot is designed to carry tools and follow you through indoor workspaces, so you can move tools or materials more efficiently while keeping your hands free for setup, inspection, and task work.

This manual explains how to prepare, operate, charge, inspect, and maintain your robot.

For best results, review the **Quick Start Guide** first, then use the later sections for detailed operating and maintenance instructions.

Environmental Constraints

Your robot performs best when you use it in a clean, predictable indoor environment with enough open floor space for smooth following and turning.

- Use your robot on level indoor flooring such as sealed concrete, tile, vinyl, laminate, or short industrial carpet.
- Keep your path reasonably clear so the robot can move beside or behind you without unnecessary stops.
- Use the robot in dry conditions and keep it away from standing water, spills, and wet cleaning zones.
- Give the robot enough turning space around benches, carts, shelves, and corners.
- Keep loose cables, rags, packaging, and small floor obstacles out of the robot's travel path.
- Operate in normal indoor lighting so you can monitor movement and respond quickly if you need to stop the robot.

System Overview

Your Mobile Tool Assistant system includes the main robot platform and the parts you need for normal operation.

Component	Description
Main robot chassis	The mobile base that carries your tool load and contains the drive system, control electronics, and frame.
Tool Platform (Payload)	The area where you place the tools or payload the robot is intended to transport.
User-tracking sensors (Robot End)	The sensing hardware used to detect and follow your position. This includes 2 UWB sensors.
User Remote	The remote that allows the robot to follow your movement. This includes a UWB sensor and a STANDBY toggle button
Obstacle detection sensors	The forward sensors used to detect nearby obstacles and trigger avoidance or stopping. This includes 3 distance sensors
Power management module	The buck converter mounted on the robot. It serves as the main power control point and provides a visual indication of battery status.
Rechargeable AA Batteries (Robot End)	The power source for the drive system and onboard electronics.

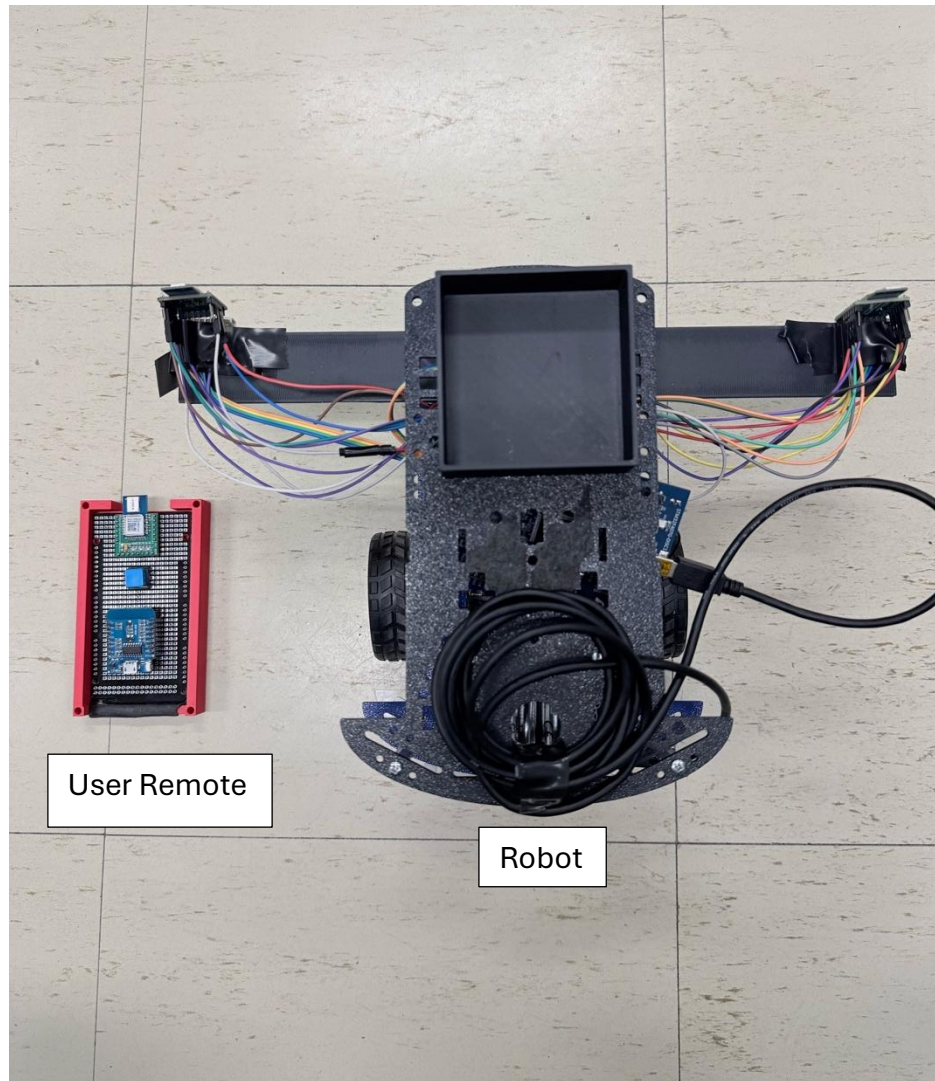


Figure 1: Top view of user remote and robot

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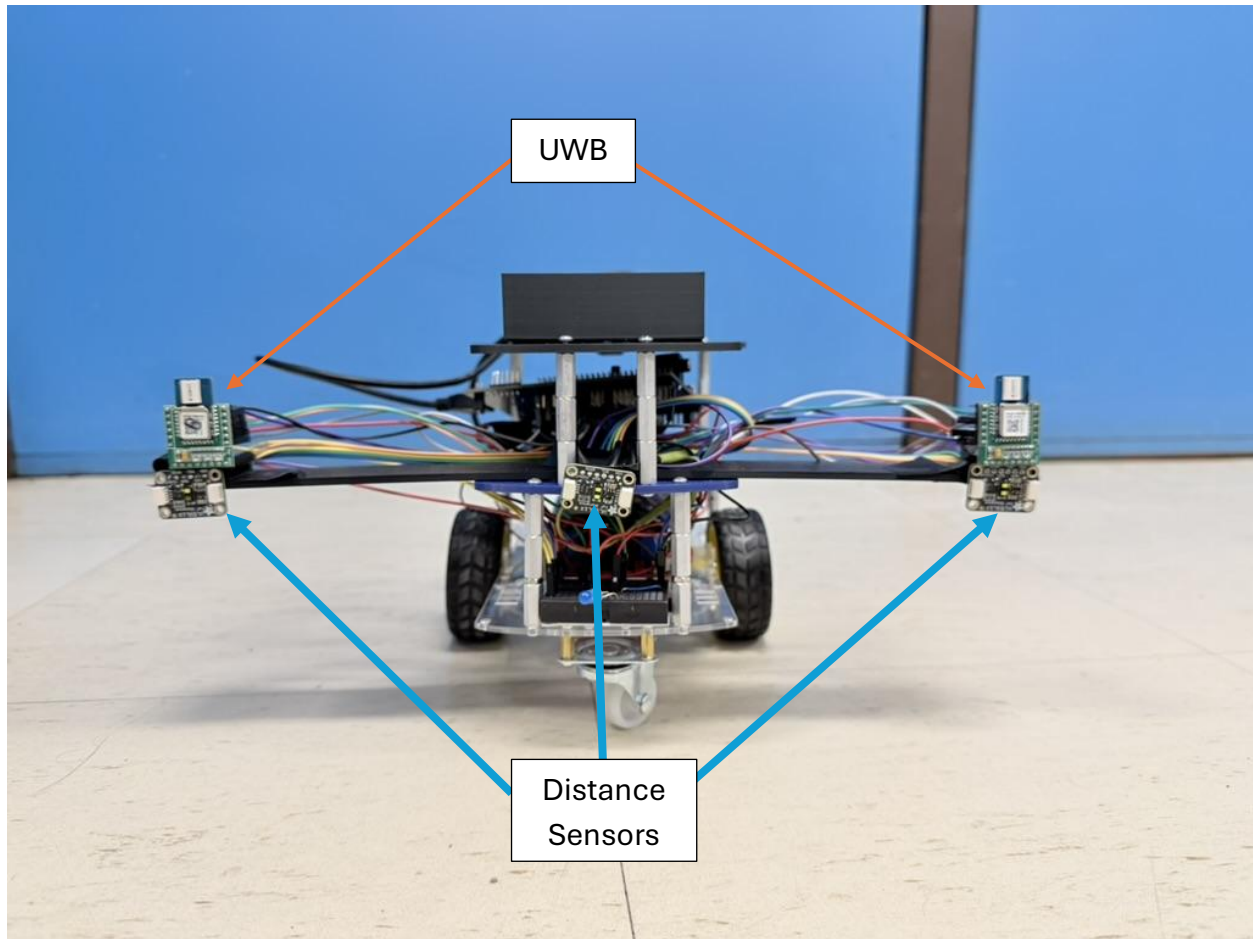


Figure 2: Front view of robot displaying sensors

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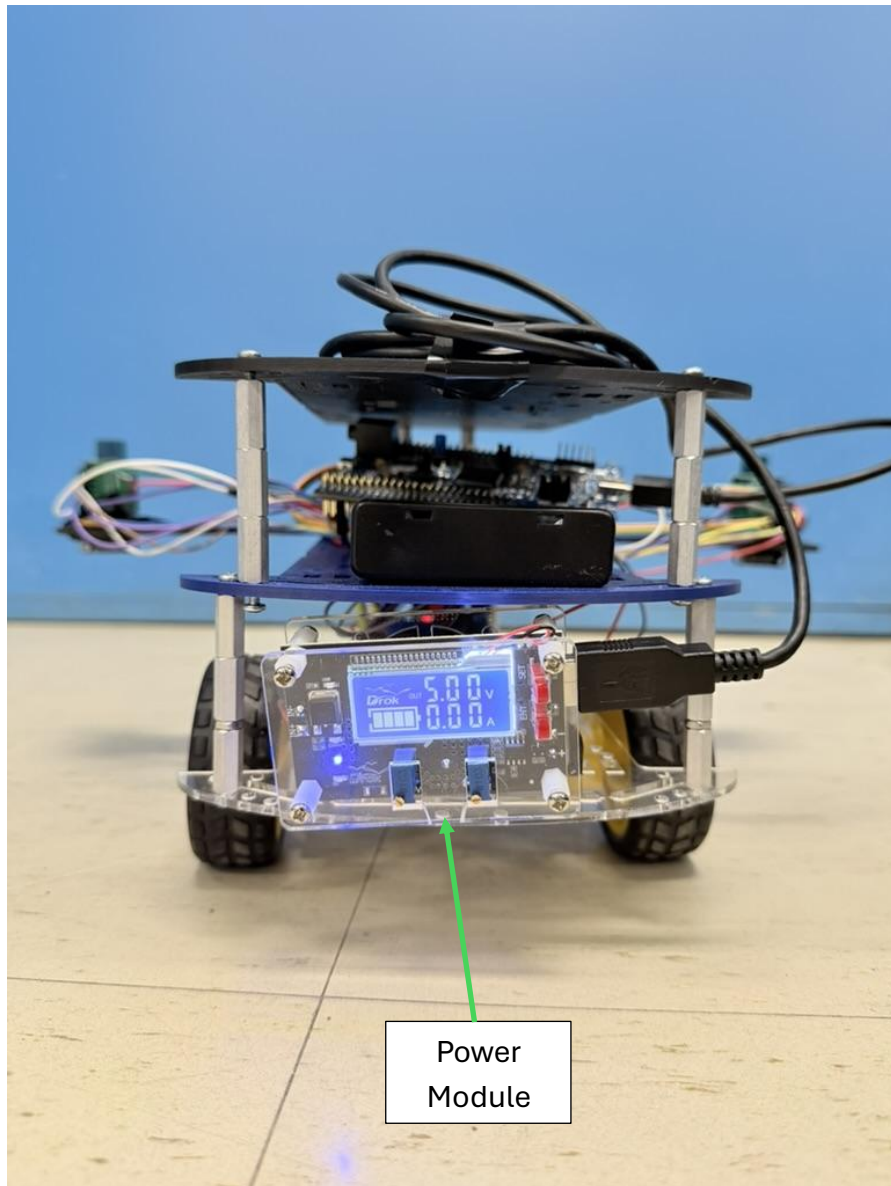


Figure 3: Back view of robot displaying power module

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Quick Start Guide

Use this section when you want the shortest path from setup to operation.

Step	Action	What you should do
1	Inspect the robot	Confirm that the robot is clean, dry, fully assembled, and free of loose wires or obvious damage.
2	Check battery charge	Make sure the battery is installed correctly and has enough charge for your session.
3	Prepare your workspace	Clear loose obstacles from the floor and confirm that your intended route is open.
4	Load your tools	Place tools securely on the platform so the load is balanced and does not hang over moving parts.
5	Power on the robot	Follow procedures explained in the Powering On section
6	Enable user tracking	Activate the tracking function by pressing the STANDBY toggle button on the User remote.
7	Begin walking	Move forward smoothly and allow the robot to establish and maintain its following distance.
8	Stop when finished	Bring the robot to a controlled stop by activating standby, and power it off, and recharge if needed.

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Operating Your Robot

Before you power on

Before each use, confirm that your battery is seated, your load is secure, your sensors are unobstructed, and the floor path is clear.

Powering on

Step 1 - Install charged batteries

Insert fully charged rechargeable AA batteries into both battery holders:

- 4 AA batteries in the upper battery tray
- 4 AA batteries in the lower battery tray

Ensure correct polarity (+/-) before proceeding.

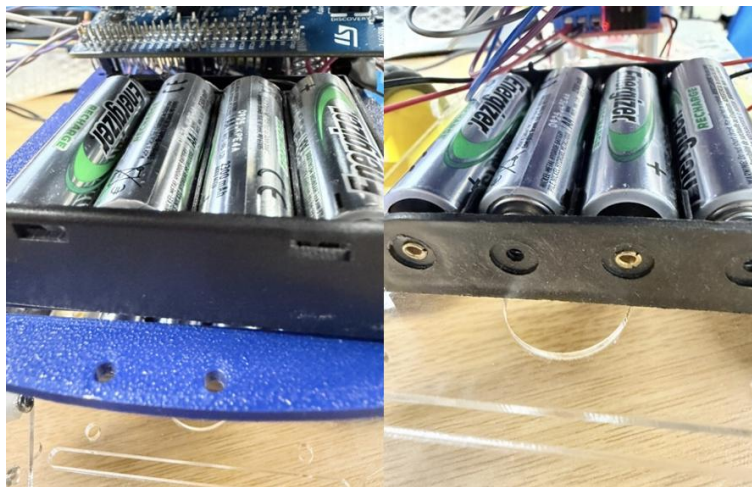


Figure 4: Upper and lower Battery tray with 4 rechargeable batteries each

Step 2 - Verify power module startup

After inserting the batteries, the power management module display will turn on.

- The display should show approximately 5.00 V output
- The charging indicator LED will be ON

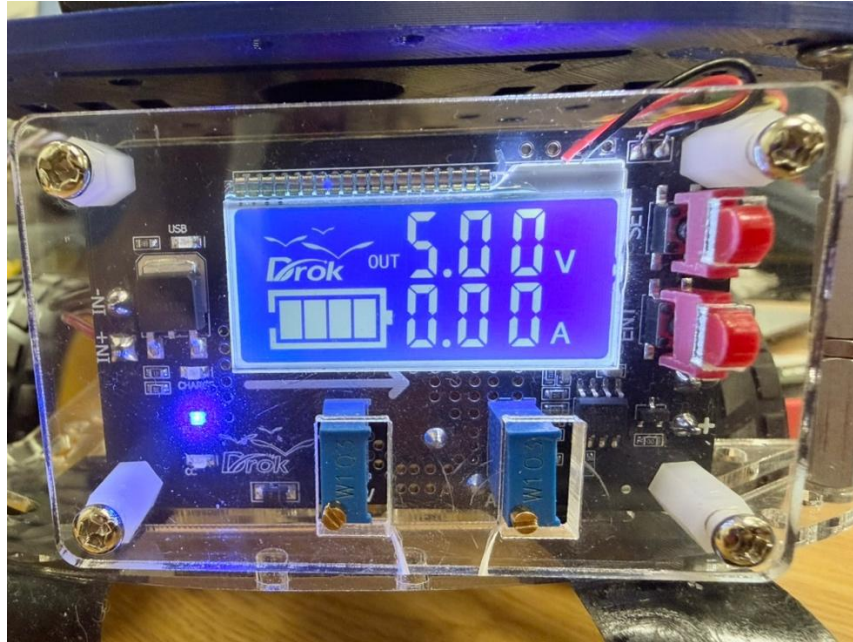


Figure 5: Power management module display after battery installation

Step 3 - Select USB output mode

Press the SET button (top red button) repeatedly until the display cycles to the USB output mode.

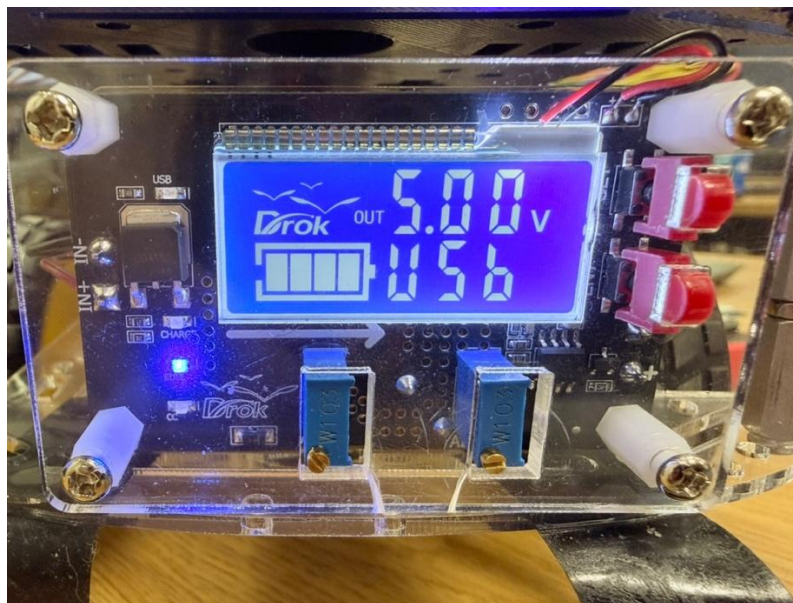


Figure 6: Cycling through power modes using the SET button to select USB output mode

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Step 4 - Enable output power

Press the ENT button (bottom red button) to enable power output.

- The USB indicator light will turn ON
- This confirms that the system is now supplying power

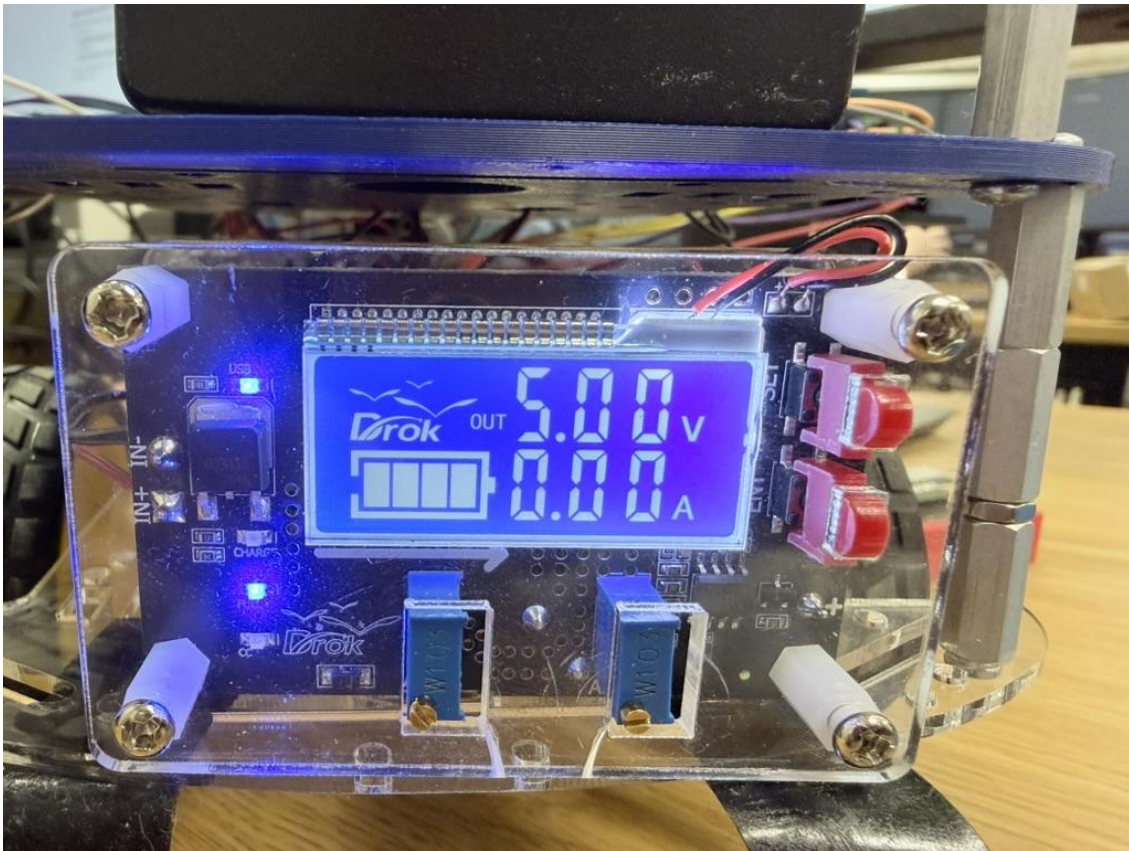


Figure 7: USB output enabled indicated by active current draw and USB indicator LED

Step 5 - Connect system power cable

Insert the USB cable connected to the robot's control board into the USB port on the power module.

- Once connected, the robot electronics will power up
- LEDs on the STM board and sensors should turn ON

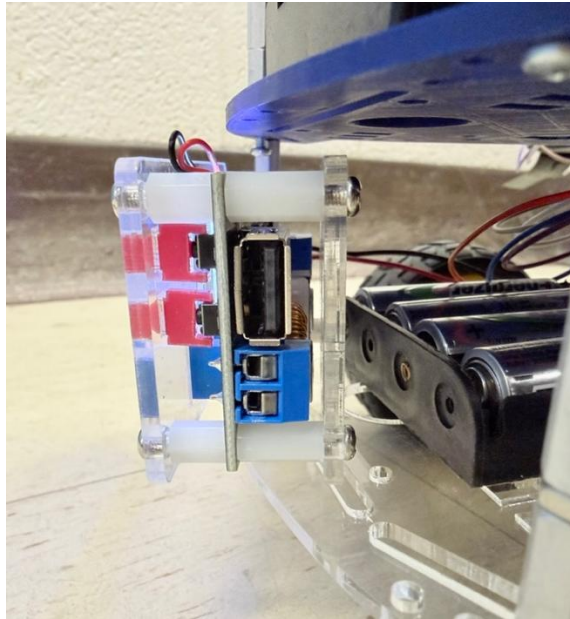


Figure 8: Connecting USB cable from robot control board to the power module

Step 6 - Confirm robot working

Verify the following before operation:

- Display shows stable voltage (~5V)
- Current draw is non-zero when system is active
- Controller and sensors are powered

The robot is now ready for operation.

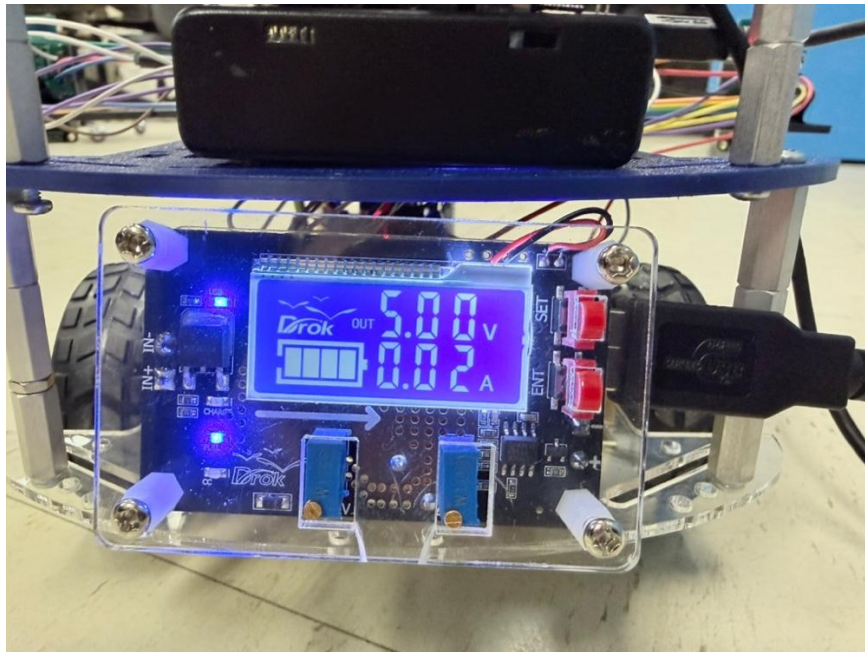


Figure 9: Robot powered on with active voltage output and current draw displayed

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Step 7 - Power the User Remote

Connect a rechargeable power bank to the User remote using a Micro-B USB cable.

- Insert the Micro-B cable into the microcontroller board on the remote
- Connect the other end to the power bank
- Confirm the remote powers on (microcontroller LED active)

The User remote is now ready for tracking operation.

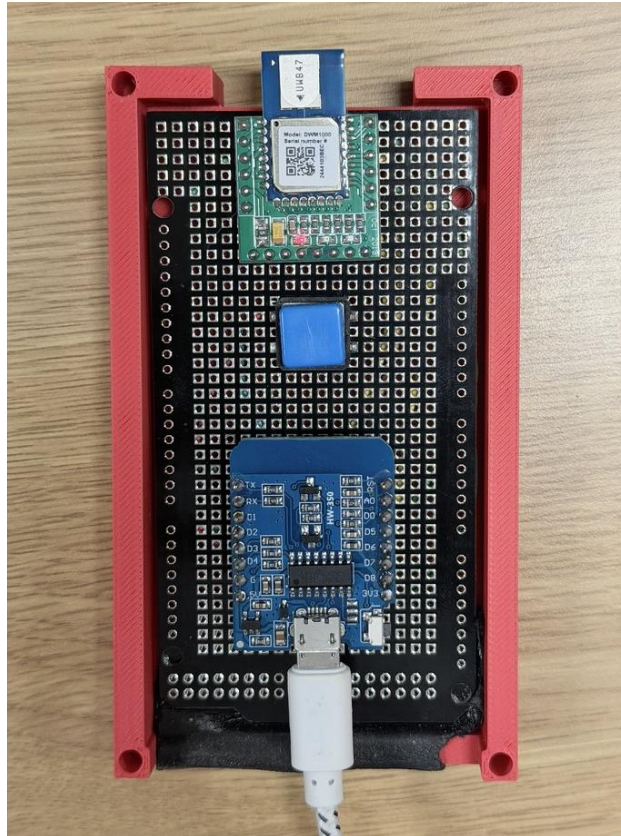


Figure 10: Connecting power to User remote using Micro-B USB cable

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Starting movement

The robot starts in standby mode by default.

To begin movement, position yourself within the robot's tracking range and field of view:

- Stand anywhere within 30 meters of the robot
- Ensure you are within the forward 180° field of view (in front of the robot)
- Face the antenna generally toward the robot when initiating tracking

Press the STANDBY button on the User remote to disable standby mode and enable user tracking.

- The blue LED on the robot will turn OFF, indicating active tracking
- Begin walking at a steady, natural pace
- Allow a few seconds for the robot to acquire your position and stabilize

Avoid sudden movement immediately after enabling tracking.

During operation

During operation, remain within the robot's tracking constraints:

- Stay within 30 meters of the robot
- Remain within the forward 180° field of view of the robot
- Avoid moving directly behind the robot without re-entering its front field of view

Walk at a consistent pace and avoid abrupt changes:

- Take wider turns around obstacles or tight spaces
- Avoid sudden stops or rapid acceleration
- Periodically check behind to ensure proper following behavior

If the robot loses tracking or behaves unexpectedly, return to its front field of view and allow it to reacquire your position.

Stopping the robot (Standby Mode)

To stop movement, place the robot into standby mode.

Press the STANDBY button on the User remote:

- The blue LED on the robot will turn ON, indicating standby mode
- The robot will immediately stop movement

While in standby mode:

- The robot remains powered
- Motors are disabled
- Pressing the STANDBY button again will resume tracking

Use standby mode whenever:

- You need to pause operation
- The robot behaves unexpectedly
- You are adjusting payload or sensors

Powering off

To fully power off the robot, disable power output from the power module.

1. Ensure the robot is in standby mode
2. Press the SET button until the display returns to the USB output setting
3. Press the ENT button to disable output - The USB indicator LED will turn OFF
4. Confirm the display remains active but output is disabled
5. Remove the batteries if required for storage or charging

The robot is now fully powered off.

Recommended operating habits

- Keep your load centered and as low as practical.
- Avoid sudden sprints, sharp pivots, or dragging the robot by hand while powered.
- Enable standby mode before adjusting wires, sensors, or the payload.
- Enable standby mode any time the robot does not respond the way you expect

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Charging and Battery Care

Charge the robot after each use or whenever the battery level is too low for reliable operation. Always ensure the robot is fully powered off before removing or installing batteries.

The robot is powered using eight (8) rechargeable AA batteries, which must be removed and charged using an external battery charger.

Use only rechargeable AA batteries (NiMH recommended) and a compatible charger. Handle batteries carefully during removal and installation to avoid damage to contacts or wiring.

The User remote is powered by a rechargeable power bank and should be charged separately as required.

Charging procedure

1. Place the robot in standby mode, then follow the powering off procedure to fully power it off.
2. Remove all 8 AA rechargeable batteries from the robot.
3. Insert the batteries into the charger, ensuring correct polarity (+/-)
4. Connect the charger to a power source
5. Confirm charging using the charger's indicator lights
6. Disconnect the User remote's power bank from the remote
7. Connect the power bank to a suitable USB Type-C charger and confirm charging
8. Wait until both the batteries and power bank are fully charged
9. Disconnect all chargers and inspect cables and connectors before next use
10. Reinstall the batteries into the robot before operation



Figure 11: AA rechargeable batteries placed in charger

Battery care guidelines

- Recharge the battery before it remains fully depleted for extended periods.
- Do not charge a battery that is visibly swollen, leaking, cracked, or overheated.
- Keep battery contacts clean and dry.
- Store batteries in a cool, dry indoor location when not in use.
- Record typical run time so you can spot battery health changes early.

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Routine Maintenance

Routine maintenance keeps your robot reliable, easier to troubleshoot, and safer to operate.

Task	Recommended frequency	What to check
Dust exterior surfaces	After each use	Remove dust and debris from the frame, tray, and exposed surfaces.
Inspect sensors	Before each use	Confirm sensors are clean and not blocked.
Check wheels and tires	Weekly	Look for debris, looseness, uneven wear, or reduced free rotation.
Inspect wiring and connectors	Weekly	Look for loose plugs, exposed conductors, pinched cables, or strain damage.
Check fasteners and mounts	Weekly	Make sure brackets, sensor mounts, and structural fasteners remain tight.
Review battery and charger condition	Weekly	Look for damage, overheating marks, or poor charge behavior.
Functional follow test	Before demonstrations	Confirm stable startup, tracking, stopping, and obstacle response.

Storage and Transport

Store your robot indoors in a clean, dry location where it will not be struck by carts, tools, or foot traffic.

Turn the robot off before storage. Recharge the batteries and power bank if the robot will not be used for an extended period.

When transporting the robot, secure it so the frame, sensors, and wheels do not shift or impact other objects. Remove or secure loose tools before moving it.

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Troubleshooting

Use the table below to diagnose common problems.

Problem	Possible cause	What you should do
Robot will not power on	Battery discharged, battery disconnected, switch off, or loose wiring	Recharge the battery, confirm all power connections, and verify that the power module is on.
Robot powers on but does not move	Tracking not enabled, start sequence incomplete, or control fault	Confirm your tracking device is active, restart the robot, and repeat the normal startup sequence.
Robot does not follow smoothly	Sensor view blocked, user position unstable, floor clutter, or uneven load	Clean the sensors, walk at a steady pace, clear the path, and rebalance the payload.
Robot stops frequently	Obstacle detection is triggered or battery is low	Check for nearby obstacles, confirm sensor cleanliness, and recharge the battery if needed.
Movement is erratic	Loose hardware, slipping wheel, control issue, or unstable sensor reading	Stop operation immediately, inspect wheels and mounts, power cycle the robot, and do a short empty-load test.
Charging does not begin	Bad charger connection, charger fault, or damaged battery	Confirm the charger is fully connected, inspect the cable and connector, and do not continue if the battery shows damage.

If a fault remains after basic checks, stop using the robot and refer the issue to the project team or a supervisor.

Technical Specifications

Overall dimensions	20 x 32 x 18cm
Robot weight	780g
Maximum payload	1kg
Battery type	Rechargeable AA NiMH batteries (1.2V)
Typical runtime	2.5 hours
Charging time	5-7 hours
Recommended operating environment	Indoor, dry, level workspace

Technical Support

Project support contact: Project team - lqv849@usask.ca , age159@usask.ca,
szj128@usask.ca , cye810@usask.ca

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